External Factors and SHAP in Urban Parking

Table of Contents

Abstract

Introduction

Purpose and Scope of the Review

- Statement of Purpose
- Specific Objectives:

Methodology of Literature Selection

- Transformation of Query
- Screening Papers
- Citation Chaining - Identifying additional relevant works
- Relevance scoring and sorting

Results

- Descriptive Summary of the Studies
- Critical Analysis and Synthesis
- Thematic Review of Literature
- Chronological Review of Literature
- Agreement and Divergence Across Studies
- Theoretical and Practical Implications
- Limitations of the Literature
- Gaps and Future Research Directions

Overall Synthesis and Conclusion

References

TLDR

This review focuses on the impact of external factors on parking demand prediction within urban parking systems, emphasizing the role of SHAP values in interpretability. It identifies key external influences such as weather, special events, and socioeconomic variables that significantly affect prediction accuracy. Findings reveal that incorporating SHAP values improves model transparency and decision-making. These insights enhance the understanding of urban parking dynamics for optimized management.

Abstract

This review synthesizes research on external factors impacting parking demand prediction, SHAP values, and urban parking systems to address gaps in model accuracy and interpretability under diverse urban conditions. The review aimed to evaluate the influence of external variables on prediction robustness, benchmark machine learning models incorporating weather, events, and urban features, analyze SHAP-based interpretability, compare spatial-temporal integration methods, and assess challenges in long-term and real-time forecasting. A systematic analysis of studies employing machine learning and deep learning techniques across global metropolitan contexts was conducted, focusing on models integrating multi-source external data and explainability tools. Key findings reveal that incorporating weather, socioeconomic, and event-related factors significantly enhances prediction accuracy and robustness, with deep learning and ensemble models outperforming traditional approaches. SHAP values improve model transparency by quantifying feature contributions, though their integration remains limited and computationally demanding. Advanced spatial-temporal architectures effectively capture complex dependencies but face scalability and heterogeneity challenges. Collectively, these findings underscore the critical role of external factors and explainability in refining urban parking demand models. The review highlights the need for holistic frameworks that integrate uncertainty quantification, dynamic external data, and scalable architectures to advance intelligent parking system design and deployment.

Introduction

Research on the external factors impacting parking demand prediction within urban parking systems has emerged as a critical area of inquiry due to its implications for urban mobility, environmental sustainability, and traffic congestion reduction(Inam et al., 2022)

(Soumana et al., 2023). Over recent years, advancements in machine learning and sensor technologies have transformed parking management, evolving from simple occupancy detection to complex predictive analytics incorporating temporal, spatial, and environmental data("Periodic Weather-Aware LSTM With Event M...", 2022)(Yang et al., 2019). The significance of this research is underscored by statistics indicating that drivers in major cities spend substantial time searching for parking, contributing to increased emissions and traffic delays(Inam et al., 2022)(Li et al., 2024). For instance, drivers in New York spend an average of 107 hours annually searching for parking, exacerbating urban congestion and pollution("Periodic Weather-Aware LSTM With Event M...", 2022)(Inam et al., 2022).

The specific problem addressed is the challenge of accurately predicting parking demand by integrating external factors such as weather, socioeconomic attributes, and urban spatial characteristics(Zhao et al., 2024)(Öz, 2023). Despite numerous predictive models, a knowledge gap persists regarding the comprehensive incorporation and interpretability of these external influences, particularly in on-street parking contexts(Inam et al., 2022)(Zhao et al., 2024)(Dahiya et al., 2024). While some studies emphasize the role of built environment and socioeconomic factors(Zhao et al., 2024)(Wang & Li, 2024), others highlight the impact of weather and policy interventions(Niu et al., 2023)(Correia et al., 2020). Controversies arise over the relative importance of these factors and the best modeling approaches, with some advocating for deep learning techniques("Parking Occupancy Prediction Method Base...", 2022)(Yin et al., 2024) and others favoring ensemble or regression models with explainability tools like SHAP values(Zhao et al., 2024)(Tanui et al., 2025). The absence of consensus limits the deployment of robust, interpretable prediction systems, potentially undermining urban traffic management and sustainability efforts(Niu et al., 2023)(Soumana et al., 2023).

Conceptually, parking demand prediction involves understanding the dynamic interplay between parking occupancy patterns, external environmental factors, and temporal-spatial dependencies("Periodic Weather-Aware LSTM With Event M...", 2022)(Zhou et al., 2024). SHAP (SHapley Additive exPlanations) values provide a framework for interpreting model outputs by quantifying feature contributions, enhancing transparency in complex models(Zhao et al., 2024)(Tanui et al., 2025). Integrating these concepts facilitates the development of predictive models that not only forecast demand but also elucidate the influence of external factors, aligning with theoretical foundations in urban informatics and machine learning interpretability(Zhao et al., 2024)(Nezhadettehad et al., 2025).

The purpose of this systematic review is to synthesize current research on the impact of external factors on parking demand prediction, emphasizing the application of SHAP values and advanced modeling techniques within urban parking systems. This review aims to bridge the identified knowledge gap by critically evaluating methodologies and interpretability approaches, thereby advancing the theoretical understanding and practical deployment of intelligent parking solutions(Zhao et al., 2024)(Tanui et al., 2025). The value lies in providing a comprehensive, evidence-based framework to guide future research and

urban planning initiatives.

This review employs a structured methodology encompassing a comprehensive literature search, inclusion of studies focusing on external factors and interpretability in parking prediction, and thematic analysis of modeling approaches and findings. The organization follows a progression from data sources and modeling techniques to interpretability methods and application contexts, facilitating a coherent synthesis of the field(Zhao et al., 2024)(Yin et al., 2024).

Purpose and Scope of the Review

Statement of Purpose

The objective of this report is to examine the existing research on "external factors impact on parking demand prediction, SHAP values, urban parking systems" in order to synthesize current knowledge on how various external influences affect the accuracy and interpretability of parking demand forecasting models within urban environments. This review is important because accurate parking demand prediction is critical for optimizing urban mobility, reducing congestion, and enhancing resource allocation in smart city infrastructures. By analyzing the role of external factors such as weather, socioeconomic attributes, events, and urban infrastructure, alongside the application of explainability techniques like SHAP values, this study aims to identify gaps, benchmark methodologies, and provide a comprehensive understanding that can guide future research and practical implementations in intelligent parking systems.

Specific Objectives:

- To evaluate current knowledge on the influence of external factors on parking demand prediction accuracy and robustness.
- Benchmarking of existing machine learning and deep learning models incorporating weather, events, and urban features for parking prediction.
- Identification and synthesis of the role of SHAP values in interpreting model predictions and external factor contributions.
- To compare methodologies addressing spatial-temporal dependencies and external data integration in urban parking systems.
- To deconstruct challenges and solutions related to long-term and real-time parking occupancy forecasting under varying external conditions.

Methodology of Literature Selection

Transformation of Query

We take your original research question — **"external factors impact on parking demand prediction, SHAP values, urban parking systems"**—and expand it into multiple, more specific search statements. By systematically expanding a broad research question into several targeted queries, we ensure that your literature search is both **comprehensive** (you won't miss niche or jargon-specific studies) and **manageable** (each query returns a set of papers tightly aligned with a particular facet of your topic).

Below were the transformed queries we formed from the original query:

- External factors impact on parking demand prediction, SHAP values, urban parking systems
- How do external factors like weather, events, and urban infrastructure influence machine learning models for predicting parking demand?
- How do external elements like weather patterns, urban infrastructure, and special events modify the accuracy of machine learning models for forecasting parking demand?

Screening Papers

We then run each of your transformed queries with the applied Inclusion & Exclusion Criteria to retrieve a focused set of candidate papers for our always expanding database of over 270 million research papers. during this process we found 129 papers

Citation Chaining - Identifying additional relevant works

- **Backward Citation Chaining:** For each of your core papers we examine its reference list to find earlier studies it draws upon. By tracing back through references, we ensure foundational work isn't overlooked.
- **Forward Citation Chaining:** We also identify newer papers that have cited each core paper, tracking how the field has built on those results. This uncovers emerging debates, replication studies, and recent methodological advances

A total of 50 additional papers are found during this process

Relevance scoring and sorting

We take our assembled pool of 179 candidate papers (129 from search queries + 50 from citation chaining) and impose a relevance ranking so that the most pertinent studies rise to the top of our final papers table. We found 178 papers that were relevant to the research

query. Out of 178 papers, 50 were highly relevant.

Results

Descriptive Summary of the Studies

This section maps the research landscape of the literature on external factors impact on parking demand prediction, SHAP values, urban parking systems, focusing on how diverse studies address the integration of external influences such as weather, socioeconomic factors, and events in forecasting urban parking demand. The reviewed works employ a variety of machine learning and deep learning methodologies, including LSTM, GRU, Transformer models, and ensemble learning, often incorporating explainability techniques like SHAP values to interpret model outputs. Geographic contexts range from large metropolitan areas in China, Singapore, and the US to European and developing cities, reflecting broad disciplinary and practical relevance. This comparative analysis elucidates the effectiveness, interpretability, and computational considerations of these models, directly informing the research questions on external factor integration and model robustness in urban parking prediction.

Study	Prediction Accuracy	External Factor Influence	Model Interpretability	Spatial-Temporal Integration	Computational Efficiency
("Periodic Weather-Aware LSTM With Event M...", 2022)	93.84% accuracy with weather and event data	Weather and events significantly improve accuracy	Moderate interpretability via event modules	Periodic patterns and weather integrated in LSTM	Efficient for large datasets, real-time feasible
(Alface et al., 2019)	Prototype model improves parking space prediction	Events, traffic, and weather integrated heuristically	Limited interpretability focus	External data fused with historical parking	Prototype with moderate computational demand
(Provoost et al., 2020)	MSE 7.18 for 60-min horizon	Weather and traffic data included	Basic interpretability via model comparison	Spatial via traffic cameras, temporal via ML	Moderate efficiency, suitable for WoT systems
(Zhang et al., 2020)	85.3% accuracy with periodic	Weather, environment, weekdays	Some interpretability through model	Periodic temporal patterns	Integrated in real system,

	weather-aware LSTM	considered	design	modeled	efficient
(Zhao et al., 2024)	Ensemble learning with SHAP reveals factor impact	Socioeconomic and built environment effects quantified	High interpretability using SHAP values	Temporal and spatial factors included	Ensemble model with moderate efficiency
(Niu et al., 2023)	Outperforms MLR, ARIMA, LSTM; policy-aware TCN	Policy, weather, time info analyzed via SHAP	High interpretability with SHAP	Temporal convolution captures policy effects	Efficient temporal convolutional network
(Zhang et al., 2024)	Improved accuracy via multifaceted spatio-temporal fusion	Holidays, POIs, weather embedded	Moderate interpretability via attention modules	Strong spatial-temporal attention mechanisms	Computationally intensive but scalable
(Fokker et al., 2021)	SARIMAX with exogenous regressors outperforms others	Weather and event factors critical	Limited interpretability	Temporal forecasting with exogenous inputs	Moderate computational cost
(Gao et al., 2023)	MAPE 2.12% outperforming LSTM and GRU	Weather and temporal factors used	Limited interpretability	Spatial-temporal via ConvGRU and wavelet transform	Efficient with denoising and feature reuse
(Wan et al., 2023)	MSE 0.0061 with attention-enhanced TCN	Dates, extreme weather, human control included	Attention mechanism aids interpretability	Temporal convolution with spatial attention	Efficient for short-term prediction
(Yang et al., 2019)	MAPE 10.6% with GCNN and LSTM	Weather and traffic speed improve accuracy	Moderate interpretability via model structure	Spatial via GCNN, temporal via LSTM	Moderate efficiency for real-time use
(Nezhadettehad et al., 2025)	27.4% accuracy improvement with Bayesian NN	Contextual features enhance uncertainty modeling	High interpretability with uncertainty quantification	Temporal and environmental factors modeled	Computationally intensive due to Bayesian inference
("Parking Occupancy Prediction	Improved accuracy and reduced time	Multi-factor input including	Moderate	Temporal dependencies	Efficient hybrid

Method Base...", 2022)	with stacked GRU-LSTM	weather and holidays	interpretability	modeled	model
(Kuhail et al., 2019)	Time-series and ML models compared for accuracy	Weather and traffic conditions cross-validated	Limited interpretability focus	Temporal forecasting with multivariate inputs	Moderate computational demand
(Yuecheng et al., 2018)	Real-time availability estimated using LSTM and external data	Meteorology, events, map mobility, POIs integrated	Moderate interpretability via model components	Temporal closeness, period, and general influence modeled	Efficient for real-time deployment
(Alajali et al., 2017)	GBRT model with 0.0291 error	Multi-source data including pedestrian and traffic	Moderate interpretability via GBRT	Temporal and spatial data fusion	Efficient ensemble method
(Öz, 2023)	Regression analysis shows weather effects on parking variables	Detailed weather categories analyzed	Limited interpretability	Temporal weather impact studied	Computationally light regression
(Canli & Toklu, 2021)	GRU model achieves 99.11% accuracy	Weather data included	Moderate interpretability	Temporal dependencies modeled	Efficient deep learning model
(Rodrigues et al., n.d.)	Classification and regression models >85% accuracy	Weather conditions integrated	Moderate interpretability	Temporal and parking features combined	Moderate computational cost
(Sun et al., 2023)	48.9% accuracy improvement with parking zone division	Spatial partitioning based on urban preferences	Moderate interpretability	Spatial graph partitioning with temporal models	Significantly improved training efficiency
(ZHANG et al., n.d.)	Multivariate linear regression for parking demand	Urban land use and traffic flow factors	Limited interpretability	Spatial factors via GIS	Computationally efficient
	BP neural				

(XIONG et al., n.d.)	network predicts parking generation rate	Socioeconomic and land use factors	Limited interpretability	Temporal forecasting	Moderate computational demand
(Zi-lin & Li-jie, n.d.)	Analysis of forecasting models for parking demand	Various models compared	Limited interpretability	Temporal focus	Varies by model
(Nasser & Simon, 2025)	MI-SMHA model reduces traffic prediction errors	Weather integrated via attention mechanisms	Moderate interpretability	Temporal and weather data fusion	Efficient sequential multihead attention
(Correia et al., 2020)	MLP predicts public transport demand with ~8% error reduction	Weather conditions two hours prior used	Limited interpretability	Temporal weather impact	Computationally light
(Wang & Li, 2024)	GBDT-SHAP reveals non-linear built environment effects	Land use mix and CBD proximity analyzed	High interpretability with SHAP	Spatial built environment factors	Efficient ensemble method
(Tanui et al., 2025)	Enhanced MLR with ROLS achieves 91.56% accuracy	Historical occupancy, temporal, weather factors	High interpretability via SHAP	Temporal and environmental features	Efficient regression with outlier handling
(Ye et al., 2023)	LSTM with temporal pattern attention improves accuracy 9.14%	Temporal decomposition and SHAP used	High interpretability with attention and SHAP	Temporal pattern attention integrated	Moderate computational cost
(Inam et al., 2022)	RF achieves 81% accuracy with multi-source data	Weather, pedestrian, traffic data impact	Moderate interpretability	Spatial and temporal data fusion	Efficient for IoT big data
	Transformer			Strong	

(Yin et al., 2024)	model outperforms others in MSE, MAE, MAPE	Multi-source traffic demand data fused	Moderate interpretability	spatial-temporal integration	Computationally intensive Transformer
(Zhang et al., 2024)	DeepPA reduces prediction error by 9.2%	Spatial and temporal factors included	Moderate interpretability	Spatial-temporal deep learning	Efficient for large-scale deployment
(Shao et al., 2024)	3% error rate with clustering and LSTM regression	Contextual features transferred across areas	Moderate interpretability	Spatial clustering and temporal modeling	Efficient transfer learning
(Lyu et al., 2024)	LSTM with POI selection reduces MAPE to 11.78%	POI data importance analyzed	Moderate interpretability	Temporal and spatial POI features	Efficient feature selection
(Balmer et al., 2021)	RF and ANN improved by geospatial info up to 25%	Land use and POI data critical	Moderate interpretability	Spatial context integrated	Moderate computational cost
(Zhao & Zhang, 2024)	AGCRU model achieves low MAE, RMSE, MAPE	POIs and household data included	Moderate interpretability	Adaptive GCN and GRU for spatial-temporal	Efficient deep learning model
(Zhou et al., 2024)	MGCN-Transformer achieves 95.93% R2	Geographic spatial and temporal features	Moderate interpretability	Multi-graph convolution and Transformer	Computationally intensive but accurate
(Li et al., 2024)	Physics-informed NN predicts occupancy and price elasticity	Economic factors integrated	Moderate interpretability	Temporal and price elasticity modeled	Moderate computational cost
(Channamallu et al., 2024)	Random forest best in regression and classification	Day of week, seasonality, time of day important	Moderate interpretability	Temporal patterns analyzed	Efficient random forest model
(Soumana et al., 2023)	Random Forest and Extra Tree best for large-scale	Weather data improves performance	Moderate interpretability	Temporal and spatial data used	Random Forest more efficient

	prediction				
(Dahiya et al., 2024)	RF outperforms KNN, SVM, DT in IoT parking prediction	Weather and environmental factors	Moderate interpretability	Temporal and spatial features	Efficient for IoT environments
(Enríquez-Aguilera et al., 2024)	AdaBoost in fog computing reduces latency	Real-world parking flow data used	Moderate interpretability	Temporal prediction	Efficient fog-cloud hybrid system
(Yang et al., 2019)	GCNN and LSTM achieve MAPE 10.6%	Weather and traffic speed improve accuracy	Moderate interpretability	Spatial and temporal data fusion	Moderate computational cost
(Shao et al., 2021)	Adversarial domain adaptation for data-scarce areas	Weather and POIs included	Moderate interpretability	Spatial-temporal contextual info	Efficient transfer learning
(Ye et al., 2023)	HGLT reduces error by 30.14% with attention	Global and local spatial views used	High interpretability via attention	Spatial and temporal attention integrated	Moderate computational cost
("Prediction of On-Street Parking Level of...", 2022)	Random Undersampling Boost classifiers improve recall	Temporal and spatial features analyzed	Moderate interpretability	Temporal mid-term data most relevant	Efficient classification model
(Chang et al., 2022)	Framework improves shared-transport user prediction	Weather, events, online activity included	Limited interpretability	Spatial, temporal, external factors fused	Moderate computational demand
(Lira et al., 2021)	ML predicts parking duration for EV demand response	Contextual and time-of-day features	Moderate interpretability	Temporal and contextual features	Efficient supervised learning

Prediction Accuracy:

- 35 studies demonstrated high prediction accuracy, often exceeding 80%, with some models achieving above 90% by integrating external factors such as weather, events,

and socioeconomic data ("Periodic Weather-Aware LSTM With Event M...", 2022) (Zhao et al., 2024) (Canli & Toklu, 2021).

- Deep learning models like LSTM, GRU, and Transformer variants consistently outperformed traditional statistical models, especially for short- and medium-term forecasts (Wan et al., 2023) (Yin et al., 2024) (Zhou et al., 2024).
- Ensemble methods such as Random Forest and Gradient Boosting also showed strong accuracy, particularly in large-scale or IoT-enabled environments (Tanui et al., 2025) (Soumana et al., 2023) (Dahiya et al., 2024).

External Factor Influence:

- 20 studies explicitly quantified the impact of weather, events, socioeconomic, and policy factors, often using SHAP values or similar explainability tools to reveal nonlinear and threshold effects (Zhao et al., 2024) (Niu et al., 2023) (Wang & Li, 2024).
- Weather variables like temperature, precipitation, and sunshine were frequently identified as significant predictors influencing parking demand and occupancy patterns (Öz, 2023) (Correia et al., 2020) (Balmer et al., 2021).
- Socioeconomic attributes such as GDP, population density, and parking pricing showed strong influence on parking utilization, with complex nonlinear relationships (Zhao et al., 2024) (Wang & Li, 2024).

Model Interpretability:

- 15 studies employed SHAP values or attention mechanisms to enhance interpretability, providing transparent explanations of feature contributions and model decisions (Zhao et al., 2024) (Niu et al., 2023) (Ye et al., 2023).
- Bayesian neural networks and physics-informed neural networks offered uncertainty quantification and economic interpretability, respectively, improving trustworthiness (Nezhadettehad et al., 2025) (Li et al., 2024).
- Some studies focused on moderate interpretability through model architecture design or feature importance without advanced explainability tools (Yang et al., 2019) (Tanui et al., 2025).

Spatial-Temporal Integration:

- 30 studies incorporated spatial dependencies and temporal patterns, using graph convolutional networks, attention mechanisms, or temporal convolutional networks to capture complex correlations (Zhang et al., 2024) (Yang et al., 2019) (Yin et al., 2024).
- Several models explicitly fused multi-source data including POIs, traffic demand, and events to enhance spatial-temporal forecasting (Inam et al., 2022) (Lyu et al., 2024) (Zhao & Zhang, 2024).
- Periodic and aperiodic temporal patterns were modeled using LSTM variants and Transformer architectures, often combined with spatial graph structures ("Periodic Weather-Aware LSTM With Event M...", 2022) (Zhang et al., 2020) (Ye et al., 2023).

Computational Efficiency:

- 18 studies reported efficient training and inference suitable for real-time or large-scale urban systems, often through model simplifications, zone partitioning, or hybrid fog-cloud architectures (Yuecheng et al., 2018) (Sun et al., 2023) (Enríquez-Aguilera et al., 2024).
- Bayesian and Transformer-based models, while accurate, tended to be more computationally intensive, requiring trade-offs for deployment (Nezhadettehad et al., 2025) (Yin et al., 2024) (Zhou et al., 2024).
- Ensemble and regression-based models generally offered faster computation with acceptable accuracy, making them practical for IoT and resource-constrained environments (Tanui et al., 2025) (Dahiya et al., 2024).

Critical Analysis and Synthesis

The reviewed literature on the impact of external factors on parking demand prediction and the use of SHAP values in urban parking systems reveals a robust engagement with diverse data sources and advanced machine learning techniques. Many studies effectively integrate weather, events, socioeconomic, and spatial-temporal data to enhance prediction accuracy and interpretability. However, challenges remain in model generalizability, real-time applicability, and the comprehensive incorporation of uncertainty and spatial heterogeneity. The use of SHAP values has improved model explainability, yet its adoption is still limited and often lacks integration with dynamic urban contexts. Overall, while methodological advancements are notable, there is a need for more holistic frameworks that address data scarcity, model transparency, and the complex interplay of external factors across different urban settings.

Aspect	Strengths	Weaknesses
Integration of External Factors	Numerous studies successfully incorporate diverse external factors such as weather, events, socioeconomic attributes, and urban infrastructure into parking demand models, demonstrating improved prediction accuracy and robustness. For example, models like PewLSTM and P-TCN effectively integrate weather and event data to capture periodic and policy-driven variations in parking behavior("Periodic Weather-Aware LSTM With Event M...", 2022)(Niu et al., 2023). Socioeconomic factors have been quantitatively analyzed using ensemble learning and SHAP to reveal their significant influence on parking utilization(Zhao et al., 2024).	Despite these integrations, many models rely heavily on historical data from limited geographic areas, which may reduce generalizability. The dynamic and nonlinear effects of external factors, such as the nonlinear impact of CBD proximity or the complex influence of holidays, are not always fully captured or modeled, limiting predictive performance in diverse urban contexts(Zhao et al., 2024)(Öz, 2023). Additionally, some studies overlook the temporal evolution of external factors, such as changing urban infrastructure or evolving socioeconomic conditions.
		The adoption of SHAP remains

Use of Explainability Techniques (SHAP Values)	The application of SHAP values in several studies provides valuable insights into feature importance and model interpretability, enhancing transparency in complex machine learning models(Zhao et al., 2024)(Niu et al., 2023)(Wang & Li, 2024). SHAP has been effectively used to quantify the contributions of socioeconomic and built environment variables, enabling urban planners to understand the mechanisms driving parking demand(Zhao et al., 2024)(Wang & Li, 2024).	relatively sparse and often limited to post-hoc analysis rather than integrated model design. Some studies do not fully exploit SHAP's potential for real-time interpretability or for guiding adaptive model updates in response to changing external conditions(Niu et al., 2023). Furthermore, the computational overhead of SHAP in large-scale or real-time systems is not thoroughly addressed, which may hinder practical deployment.
Modeling Spatial-Temporal Dependencies	Advanced deep learning architectures, including LSTM variants, graph convolutional networks, and Transformer-based models, have been employed to capture complex spatial-temporal correlations in parking data(Zhang et al., 2024)(Yang et al., 2019)(Yin et al., 2024)(Zhou et al., 2024). These models demonstrate superior performance by integrating multi-source data and effectively modeling periodicity and spatial heterogeneity(Zhang et al., 2024)(Yin et al., 2024).	Despite these advances, challenges persist in scaling models to large urban networks due to computational costs and data sparsity. Some approaches, such as parking zone division, attempt to mitigate this but may introduce biases or overlook fine-grained spatial interactions(Sun et al., 2023). Moreover, the heterogeneity of parking types (on-street vs. off-street) and their distinct spatial dynamics are not always adequately differentiated in modeling frameworks(Balmer et al., 2021).
Handling of Uncertainty and Data Scarcity	Innovative approaches like Bayesian Neural Networks explicitly model uncertainty, improving robustness under data scarcity and noise, which is critical for real-world deployment(Nezhadettehad et al., 2025). Transfer learning and adversarial domain adaptation frameworks address the challenge of insufficient local data by leveraging information from similar areas(Shao et al., 2024)(Shao et al., 2021).	However, uncertainty modeling is still underrepresented in the literature, with most studies focusing on point predictions without confidence quantification(Nezhadettehad et al., 2025). Transfer learning methods may suffer from domain mismatch and require careful feature alignment, which is not always systematically addressed(Shao et al., 2021). Additionally, few studies explore uncertainty-aware decision-making in parking management.
Comparative Evaluation of Machine	Several works conduct comprehensive benchmarking of machine learning algorithms, revealing that ensemble methods like Random Forest often outperform more complex deep learning models in terms of accuracy and computational efficiency(Inam et al., 2022)	Many comparative studies are limited by the scope of datasets, often focusing on single cities or specific parking types, which restricts the generalizability of conclusions(Inam et al., 2022)(Soumana et al., 2023). Furthermore, evaluation metrics vary

Learning Models	(Soumana et al., 2023)(Dahiya et al., 2024). Hybrid models combining GRU and LSTM have shown improvements in both prediction accuracy and efficiency("Parking Occupancy Prediction Method Base...", 2022)(Zeng et al., n.d.).	widely, complicating cross-study comparisons. The lack of standardized benchmarks and real-time performance assessments limits the practical applicability of reported results.
Impact of External Factors on Prediction Accuracy and Interpretability	Empirical evidence consistently shows that incorporating external factors such as weather, events, and socioeconomic variables significantly enhances prediction accuracy and model interpretability("Periodic Weather-Aware LSTM With Event M...", 2022)(Zhao et al., 2024)(Öz, 2023)(Correia et al., 2020). Studies highlight the critical role of event-related variables and weather conditions in capturing demand fluctuations(Fokker et al., 2021)(Öz, 2023).	Nonetheless, the interaction effects among external factors are often insufficiently modeled, leading to potential oversimplifications(Zhao et al., 2024). Some studies report that the influence of weather and events varies by location and time horizon, indicating a need for adaptive models that can dynamically weigh external inputs(Balmer et al., 2021). Additionally, the temporal granularity of external data is sometimes coarse, limiting responsiveness to sudden changes.
Real-Time Applicability and Scalability	Models integrated into real-time systems, such as Du-Parking and fog computing architectures, demonstrate practical feasibility and reduced latency, enhancing user experience and operational efficiency(Yuecheng et al., 2018)(Enríquez-Aguilera et al., 2024). Zone-based partitioning and parallel training approaches improve scalability for large urban networks(Sun et al., 2023).	Despite these advances, many models remain computationally intensive and require extensive data preprocessing, which may impede real-time deployment in resource-constrained environments(Sun et al., 2023)(Enríquez-Aguilera et al., 2024). The integration of real-time external data streams, such as live weather or event updates, is not uniformly addressed, limiting responsiveness. Moreover, few studies explore the trade-offs between model complexity and latency in operational settings.

Thematic Review of Literature

Research on parking demand prediction and urban parking systems consistently highlights the critical influence of external factors such as weather, socioeconomic attributes, events, and urban infrastructure on model accuracy and interpretability. Machine learning models, particularly deep learning architectures like LSTM, GRU, and Transformer-based networks, have been extensively developed to incorporate these factors for improved forecasting. The integration of explainability techniques, especially SHAP values, is increasingly emphasized to elucidate the contribution of external variables, enhancing model transparency and decision support. Spatial-temporal dependencies, data fusion from

heterogeneous sources, and the handling of real-time versus long-term prediction challenges form core methodological pillars evolving alongside the increasing availability of large-scale urban parking datasets.

Theme	Appears In	Theme Description
Influence of External Factors (Weather, Socioeconomic, Events) on Parking Demand Prediction	30/50 Papers	Extensive research underscores the significant effects of weather conditions, socioeconomic variables, and event-related factors on parking occupancy prediction accuracy. Studies demonstrate that weather features like temperature, precipitation, and sunshine, alongside socioeconomic indicators such as GDP and population density, shape parking demand patterns nonlinearly, necessitating their integration into predictive models to enhance robustness and relevance ("Periodic Weather-Aware LSTM With Event M...", 2022) (Zhao et al., 2024) (Niu et al., 2023) (Öz, 2023) (Balmer et al., 2021) (Chang et al., 2022). The role of events and holidays is also critical, affecting temporal dynamics and requiring dedicated modeling components ("Periodic Weather-Aware LSTM With Event M...", 2022) (Fokker et al., 2021).
Machine Learning and Deep Learning Models for Parking Prediction	28/50 Papers	A wide array of machine learning and deep learning models—LSTM, GRU, Transformer, Random Forest, and Graph Convolutional Networks—have been developed to forecast parking space availability. Hybrid models combining temporal and spatial features and incorporating external data sources have achieved superior predictive accuracy and efficiency. Deep learning architectures, such as periodic weather-aware LSTM and multi-graph convolutional Transformer, account for complex nonlinearities and spatio-temporal correlations in parking data ("Periodic Weather-Aware LSTM With Event M...", 2022) (Provoost et al., 2020) ("Parking Occupancy Prediction Method Base...", 2022) (Yin et al., 2024) (Zhou et al., 2024) (Yang et al., 2019). Comparative studies often highlight random forest and ensemble methods for their balance of performance and computational demand in large-scale scenarios (Inam et al., 2022) (Soumana et al., 2023) (Dahiya et al., 2024).
Application of SHAP Values and Explainability Techniques	10/50 Papers	SHAP values are increasingly utilized to interpret model outputs, revealing the relative importance and nonlinear effects of external factors on parking demand predictions. This approach provides transparency in complex models, enabling stakeholders to understand feature contributions and decision boundaries, which is crucial for practical deployment in smart city infrastructures (Zhao et al., 2024) (Niu et al., 2023) (Tanui et al., 2025) (Ye et al., 2023). SHAP-based analyses have identified key predictors such as economic attributes, weather parameters, and policy-related features, facilitating targeted interventions and pricing strategies (Niu et al., 2023) (Wang & Li, 2024) (Li et al., 2024).
		Effective parking demand forecasting requires capturing both spatial relationships among parking lots and temporal patterns, including periodicity and irregular events. Advanced models employ graph neural

Integration of Spatial-Temporal Dependencies and Multi-Source Data Fusion	18/50 Papers	networks, attention mechanisms, and clustering techniques to capture these dependencies. Multi-source data fusion, combining parking occupancy, traffic flow, weather, POIs, and socioeconomic data, enhances prediction accuracy and system scalability (Zhang et al., 2024) (Yang et al., 2019) (Yuecheng et al., 2018) (Inam et al., 2022) (Yin et al., 2024) (Zhang et al., 2024). The spatial heterogeneity and temporal variability of urban parking demand necessitate adaptive frameworks capable of handling large-scale and heterogeneous data efficiently (Sun et al., 2023) (Shao et al., 2024).
Challenges and Solutions in Real-Time and Long-Term Parking Forecasting	12/50 Papers	Predicting parking occupancy across different time horizons poses challenges related to data scarcity, noise, and concept drift, especially under dynamic urban conditions such as pandemics or policy changes. Models incorporating uncertainty quantification, such as Bayesian neural networks, and policy-aware architectures improve reliability and adaptability. Real-time systems balance accuracy with computational efficiency, often leveraging transfer learning and domain adaptation to overcome sparse data in new areas (Niu et al., 2023) (Nezhadettehad et al., 2025) (Yuecheng et al., 2018) (Shao et al., 2021) (Shao et al., 2020). Long-term forecasting benefits from integrating periodic and event-driven features to capture evolving urban mobility patterns (Zhang et al., 2020) (Fokker et al., 2021).
Urban Infrastructure and Built Environment Effects on Parking Demand	8/50 Papers	The built environment, including land use mix, proximity to CBDs, and mechanical parking lot planning, significantly influences parking demand patterns. Studies utilizing gradient boosting and other interpretable models reveal nonlinear relationships between urban form and parking space utilization, providing actionable insights for planners (Zhao et al., 2024) (Wang & Li, 2024) (Balmer et al., 2021). Incorporation of POIs and housing data further refines spatial prediction accuracy, highlighting the contextual dependency of parking behaviors on urban morphology (Lyu et al., 2024) (Zhao & Zhang, 2024).
Data-Driven Large-Scale and Cross-Domain Parking Prediction	7/50 Papers	Large-scale datasets from diverse urban contexts, including Singapore and multiple Chinese cities, enable the development of scalable and transferable parking prediction models. Novel datasets combining cross-domain factors and advanced graph-based operators address computational complexity while maintaining high accuracy. Transfer learning and few-shot domain adaptation frameworks facilitate model generalization to data-scarce regions (Zhang et al., 2024) (Zhang et al., 2024) (Shao et al., 2024) (Shao et al., 2021). Deployment in practical platforms supports real-time urban mobility management and planning (Zhang et al., 2024) (Enríquez-Aguilera et al., 2024).

Chronological Review of Literature

The literature on the impact of external factors on parking demand prediction and the use of SHAP values in urban parking systems has evolved significantly over recent years. Early

research focused on integrating weather and event data with basic machine learning models to improve prediction accuracy. Over time, advancements in deep learning, spatio-temporal modeling, and multi-source data fusion have enhanced the sophistication and performance of prediction models. Recently, explainability techniques such as SHAP values and uncertainty-aware models have gained prominence, providing deeper interpretability and robustness to forecasting efforts in smart city contexts.

Year Range	Research Direction	Description
2017–2019	Foundational Integration of External Data in Parking Prediction	Initial studies focused on incorporating external factors such as weather, events, and traffic data into parking demand models using machine learning approaches. These works introduced multi-source data fusion and established the importance of spatial and temporal dependencies for improving accuracy. Early deep learning models combined LSTM and GCNN for real-time occupancy prediction, demonstrating superior performance to traditional methods.
2020–2021	Advancement in Deep Learning and Spatial-Temporal Modeling	Research progressed towards employing advanced neural network architectures such as periodic weather-aware LSTM models and hybrid approaches combining GRU and LSTM. Spatial relationships and temporal autocorrelations were modeled more effectively using graph convolutional networks and attention mechanisms. Studies began exploring the influence of socioeconomic and built environment factors, alongside weather and event variables, to better understand urban parking dynamics.
2022–2023	Enhanced Model Complexity and Explainability Techniques	The literature saw a rise in complex models integrating diverse external factors including holidays, socioeconomic attributes, and pandemic-related policies. Temporal convolutional networks and stacked GRU-LSTM hybrids were employed for short- and long-term predictions. SHAP values became a key tool for interpreting model outputs and feature contributions, while policy-aware and uncertainty-aware models addressed real-world irregularities and data noise.
2024–2025	Multi-Source Data Fusion, Transformer Models, and Proactive Management	Recent research emphasizes multi-source traffic demand data fusion with transformer architectures to capture complex spatial-temporal dependencies. Explainability via SHAP and Bayesian neural networks enhances interpretability and robustness in predictions. Emerging works focus on integrating these predictive models with proactive parking pricing strategies and large-scale IoT-enabled smart parking systems to optimize urban mobility and resource utilization.

Agreement and Divergence Across Studies

There is widespread agreement that external factors such as weather, events, socioeconomic variables, and urban infrastructure significantly influence parking demand prediction accuracy. Many studies concur that incorporating these factors alongside

spatial-temporal modeling improves predictive performance and robustness. However, divergences arise in the specific modeling approaches, data integration techniques, and the use of explainability methods like SHAP, with varying emphasis on real-time versus long-term forecasting. Differences in urban context, scale, and data availability further contribute to contrasting findings on computational efficiency and interpretability.

Comparison Criterion	Studies in Agreement	Studies in Divergence	Potential Explanations
Influence of External Factors (weather, events, socioeconomic)	Multiple studies confirm the critical role of weather, events, and socioeconomic variables in enhancing parking demand prediction accuracy and robustness ("Periodic Weather-Aware LSTM With Event M...", 2022) (Zhao et al., 2024) (Niu et al., 2023) (Öz, 2023) (Correia et al., 2020) (Inam et al., 2022) (Balmer et al., 2021). Weather and event variables notably improve model performance across contexts ("Periodic Weather-Aware LSTM With Event M...", 2022) (Fokker et al., 2021) (Yang et al., 2019). Socioeconomic attributes have been shown to sometimes surpass built environment factors in importance for parking demand (Zhao et al., 2024).	Some studies focus more narrowly on built environment or temporal features, downplaying socioeconomic or event factors (Zhang et al., 2020) (Wang & Li, 2024) (XIONG et al., n.d.). The degree of external factor impact varies, with some models emphasizing periodicity or spatial relations over external factors (Zhang et al., 2020) (Zhang et al., 2024) (Zhou et al., 2024).	Variations stem from data scope (on-street vs off-street), urban context (city size, culture), and study objectives (short-term vs long-term forecasts). Different feature engineering approaches and geographic focus influence factor importance.
Use and Effectiveness of SHAP for Model Interpretability	Studies employing SHAP consistently report enhanced interpretability and identification of key external factors influencing parking demand, such as socioeconomic attributes, weather, and pricing (Zhao et al., 2024) (Niu et al., 2023) (Wang & Li, 2024) (Tanui et al., 2025) (Ye et al., 2023). SHAP helps reveal nonlinear and threshold effects, improving transparency in ensemble and deep learning models.	Some recent deep learning studies do not incorporate SHAP or focus on other explainability approaches like attention mechanisms or domain adaptation, limiting interpretability discussion (Yin et al., 2024) (Shao et al., 2024) (Shao et al., 2021). Additionally, certain Bayesian or uncertainty-aware models highlight probabilistic interpretability over SHAP (Nezhadettehad	Differences likely arise due to model type (ML vs deep learning), computational overhead of SHAP in large-scale systems, and evolving emphasis on explainability frameworks. Studies in developing contexts may prioritize computational efficiency over

		et al., 2025).	interpretability.
Spatial-Temporal Integration with External Factors	Broad consensus exists that integrating spatial-temporal dependencies with external factors improves prediction significantly, as seen in models using graph convolution, attention mechanisms, and hybrid architectures (Zhang et al., 2024) (Yang et al., 2019) (Yin et al., 2024) (Zhao & Zhang, 2024) (Zhou et al., 2024) (Ye et al., 2023). Periodic and aperiodic temporal patterns with spatial clustering are widely recognized as important ("Periodic Weather-Aware LSTM With Event M...", 2022) (Ye et al., 2023).	Some models focus primarily on temporal or spatial features without explicit or advanced fusion of external factors ("Parking Occupancy Prediction Method Base...", 2022) (Zeng et al., n.d.) (Yuecheng et al., 2018). The granularity of spatial modeling (block-level vs parking lot clusters) varies, affecting integration depth (Sun et al., 2023) (Yang et al., 2019).	Variations in data availability, urban scale, and computational resources influence the extent and method of spatial-temporal-external factor fusion. Novel approaches like Transformers or graph-based partitioning differ in complexity and applicability.
Model Prediction Accuracy and Benchmarking	Many studies report improved accuracy through machine learning and deep learning models that incorporate external factors, with models like PewLSTM, stacked GRU-LSTM, TCN, and Transformer variants outperforming classical baselines ("Periodic Weather-Aware LSTM With Event M...", 2022) (Niu et al., 2023) (Gao et al., 2023) (Wan et al., 2023) (Yin et al., 2024) (Zhou et al., 2024). Random Forest is frequently noted for reliable performance in various contexts (Inam et al., 2022) (Channamallu et al., 2024) (Soumana et al., 2023) (Dahiya et al., 2024).	Some divergence exists in best-performing models depending on prediction horizon and context, with complex models sometimes outperformed by simpler ensemble methods in terms of accuracy or computational efficiency (Inam et al., 2022) (Soumana et al., 2023) (Dahiya et al., 2024). Bayesian and uncertainty-aware models prioritize reliability over raw accuracy (Nezhadettehad et al., 2025).	Differences arise from dataset characteristics, urban environment complexity, model capacity, and external factors included. Trade-offs exist between accuracy, interpretability, and computational demands.
		Deep learning models with complex spatial-temporal attention or	The trade-off between model complexity and

Computational Efficiency and Scalability in Large-Scale Urban Systems	Approaches using graph partitioning, model simplification (e.g., DCT-based GCO), and ensemble methods emphasize reducing training and inference time, improving scalability (Sun et al., 2023) (Zhang et al., 2024). Random Forest and other less complex models are praised for execution speed and adaptability (Inam et al., 2022) (Soumana et al., 2023). Fog computing architectures help reduce latency in real-time systems (Enríquez-Aguilera et al., 2024).	Bayesian formulations often face higher computational costs and complexity, potentially limiting real-time deployment (Nezhadettehad et al., 2025) (Yin et al., 2024) (Zhou et al., 2024). Some studies do not address efficiency explicitly ("Periodic Weather-Aware LSTM With Event M...", 2022) (Yuecheng et al., 2018).	efficiency is influenced by computational resources, real-time system requirements, and data volume. Model design choices reflect priorities between accuracy, interpretability, and speed.
Handling of Long-term vs Real-time Forecasting Under External Influences	Studies recognize the challenges of long-term forecasting and highlight the importance of external factors and model adaptability for both horizons (Niu et al., 2023) (Fokker et al., 2021) (Nezhadettehad et al., 2025) (Zhou et al., 2024). Models incorporating event and policy awareness improve long-term prediction robustness (Niu et al., 2023).	Some research prioritizes short-term or ultra-short-term prediction accuracy with rapid updates, potentially at the expense of longer-term forecasting reliability (Wan et al., 2023) ("Parking Occupancy Prediction Method Base...", 2022) (Lyu et al., 2024). Approaches vary in temporal scope and update frequency.	Differences are due to intended application (e.g., dynamic parking guidance vs urban planning), data update frequency, and model design (sequence length, temporal granularity). Long-term forecasting demands more robust external factor integration and uncertainty modeling.

Theoretical and Practical Implications

Theoretical Implications

- The synthesis of findings underscores the critical role of external factors such as

weather, socioeconomic attributes, and events in enhancing the accuracy and robustness of parking demand prediction models. This supports and extends existing theories that emphasize the multifactorial nature of urban mobility and parking behavior, highlighting the necessity of integrating diverse data sources for improved predictive performance(Zhao et al., 2024) (Niu et al., 2023) (Öz, 2023).

- The application of explainability techniques, particularly SHAP values, has advanced the interpretability of complex machine learning models in parking demand forecasting. This development challenges traditional black-box approaches by providing transparent insights into feature importance and nonlinear effects, thereby enriching theoretical frameworks around model explainability in urban systems(Zhao et al., 2024) (Niu et al., 2023) (Tanui et al., 2025).
- The integration of spatial-temporal dependencies through advanced architectures such as graph convolutional networks, Transformers, and hybrid deep learning models confirms the theoretical importance of capturing both local and global spatial correlations alongside temporal patterns. This aligns with and expands upon spatial-temporal modeling theories in intelligent transportation systems(Yin et al., 2024) (Zhao & Zhang, 2024) (Ye et al., 2023).
- The recognition of periodic patterns and their incorporation into recurrent neural network models, such as periodic weather-aware LSTM, reinforces the theoretical understanding that temporal regularities are fundamental to parking behavior prediction. This challenges models that neglect periodicity and supports the inclusion of cyclical temporal features for enhanced forecasting(Zhang et al., 2020) ("Periodic Weather-Aware LSTM With Event M...", 2022).
- The emerging focus on uncertainty quantification through Bayesian neural networks introduces a theoretical paradigm shift by explicitly modeling epistemic and aleatoric uncertainties. This approach enhances the robustness and reliability of predictions, addressing limitations in conventional deterministic models and contributing to the theory of uncertainty-aware machine learning in urban mobility(Nezhadettehad et al., 2025).

Practical Implications

- For urban planners and policymakers, the demonstrated influence of socioeconomic factors and built environment characteristics on parking demand suggests that parking management strategies should be tailored to local economic and demographic contexts. This can inform dynamic pricing, zoning, and infrastructure development to optimize parking utilization(Zhao et al., 2024) (Wang & Li, 2024) (Li et al., 2024).
- The successful deployment of models incorporating weather, events, and policy variables in real-time or near-real-time systems highlights the practical feasibility of integrating heterogeneous data streams into smart parking platforms. This can improve driver experience by providing accurate availability forecasts and reduce congestion caused by parking search(Zhang et al., 2020) (Niu et al., 2023) (Yuecheng

et al., 2018).

- The use of explainability methods like SHAP in operational models enhances stakeholder trust and facilitates transparent decision-making in parking management systems. This is particularly valuable for justifying pricing policies, resource allocation, and adaptive control measures in smart city contexts(Zhao et al., 2024) (Tanui et al., 2025).
- The advancement of scalable and efficient computational methods, such as parking zone division and graph cosine operators, addresses practical challenges related to large-scale urban parking networks. These innovations enable faster model training and deployment, making citywide parking prediction more accessible and actionable(Sun et al., 2023) (Zhang et al., 2024).
- The integration of multi-source data fusion, including traffic demand from various transportation modes, POIs, and spatial features, supports the development of comprehensive parking prediction frameworks that can adapt to complex urban dynamics. This facilitates more effective parking guidance and urban mobility planning(Yin et al., 2024) (Lyu et al., 2024) (Zhao & Zhang, 2024).
- The incorporation of uncertainty-aware models and adaptive learning frameworks in parking prediction systems offers practical benefits in handling data scarcity, noise, and changing urban conditions. This enhances the resilience and reliability of parking management solutions, particularly in rapidly evolving or resource-constrained environments(Nezhadettehad et al., 2025) (Shao et al., 2021).

Limitations of the Literature

Area of Limitation	Description of Limitation	Papers which have limitation
Geographic Bias	Many studies focus on specific cities or regions, limiting the generalizability of findings to other urban contexts with different infrastructure, cultural, or socioeconomic characteristics. This geographic concentration reduces external validity and applicability across diverse urban settings.	("Periodic Weather-Aware LSTM With Event M...", 2022) (Zhang et al., 2020) (Zhao et al., 2024) (Canli & Toklu, 2021) (Balmer et al., 2021) (Zhao & Zhang, 2024) (Zhang et al., 2024) (Zhang et al., 2024)
Limited Real-Time Data Integration	Several models rely heavily on historical data without sufficient incorporation of real-time or dynamic data streams, which constrains their ability to adapt to sudden changes in parking demand caused by events or emergencies, thus affecting model robustness and practical deployment.	(Alface et al., 2019) (Fokker et al., 2021) (Yuecheng et al., 2018) (Yang et al., 2019)
	While external factors like weather, events,	

Insufficient Handling of External Factors	and socioeconomic attributes are recognized, many models inadequately capture their complex, nonlinear, and dynamic influences on parking demand, limiting interpretability and prediction accuracy under varying conditions.	(Zhao et al., 2024) (Niu et al., 2023) (Öz, 2023) (Ye et al., 2023) (Lyu et al., 2024) (Balmer et al., 2021) (Chang et al., 2022)
Model Complexity vs. Interpretability	Advanced deep learning and ensemble models often achieve high accuracy but at the cost of interpretability, making it difficult to understand the contribution of external factors and limiting transparency for decision-makers and stakeholders.	(Nezhadettehad et al., 2025) (Tanui et al., 2025) (Shao et al., 2024) (Shao et al., 2021) (Shao et al., 2020)
Data Scarcity and Transferability Issues	Many approaches require extensive historical data, which is unavailable in new or rapidly changing urban areas, hindering model applicability and necessitating transfer learning or domain adaptation techniques that are still underdeveloped.	(Shao et al., 2024) (Shao et al., 2021) (Shao et al., 2020)
Short Prediction Horizons	Numerous studies focus on short-term or ultra-short-term parking demand predictions, which may not adequately support long-term planning and policy-making, thereby limiting strategic urban mobility management.	(Fokker et al., 2021) (Wan et al., 2023) (Ye et al., 2023) (Yang et al., 2019)
Limited Consideration of Spatial-Temporal Dependencies	Some models inadequately address complex spatial-temporal correlations and interactions among parking lots, which are critical for capturing realistic parking dynamics and improving prediction accuracy.	(Zhang et al., 2024) (Yang et al., 2019) (Yin et al., 2024) (Zhou et al., 2024) (Ye et al., 2023)
Underrepresentation of Economic and Pricing Factors	Few studies integrate economic variables such as pricing and elasticity into parking demand models, which restricts understanding of demand responsiveness and limits the development of proactive pricing strategies.	(Zhao et al., 2024) (Li et al., 2024)
Computational Efficiency Constraints	High computational costs associated with complex models, especially those incorporating graph-based or Bayesian methods, limit scalability and real-time applicability in large urban networks, affecting practical deployment.	(Nezhadettehad et al., 2025) (Sun et al., 2023) (Zhang et al., 2024)

Gaps and Future Research Directions

Gap Area	Description	Future Research Directions	Justification	Research Priority
Limited integration of dynamic external factors	Most models incorporate static or historical external factors (weather, events, socioeconomic) but lack dynamic real-time updates and adaptive weighting of these factors in prediction models.	Develop adaptive models that integrate real-time external data streams (e.g., live weather, events, policy changes) with dynamic feature importance adjustment using SHAP or attention mechanisms.	Real-time responsiveness is critical for accurate parking demand prediction in fluctuating urban environments, yet current models often rely on static or delayed data, limiting practical applicability ("Periodic Weather-Aware LSTM With Event M...", 2022) (Niu et al., 2023) (Yuecheng et al., 2018).	High
Sparse use of SHAP for real-time interpretability and model adaptation	SHAP values are mainly used for post-hoc analysis rather than integrated into model training or real-time decision-making. Computational overhead also limits SHAP use in large-scale or real-time systems.	Investigate efficient SHAP approximations or incremental SHAP computations to enable real-time interpretability and adaptive model updates based on changing external factors.	Enhancing model transparency and trustworthiness in operational settings requires real-time explainability, which is currently underexplored and computationally challenging (Zhao et al., 2024) (Niu et al., 2023) (Ye et al., 2023).	High
Insufficient modeling of nonlinear and interaction effects among external factors	Many studies model external factors independently or linearly, neglecting complex nonlinear interactions (e.g., between socioeconomic variables and spatial proximity).	Employ advanced nonlinear modeling techniques (e.g., GBDT-SHAP, physics-informed NNs) to capture interaction effects and threshold behaviors among external factors in parking demand.	Nonlinear and interaction effects significantly influence parking utilization patterns, as shown by nonlinear CBD proximity effects and socioeconomic impacts, yet are underrepresented in models (Zhao et al., 2024) (Wang & Li, 2024) (Li et al., 2024).	High
Limited uncertainty quantification in parking demand predictions	Most models provide point estimates without quantifying prediction uncertainty, limiting robustness and decision-making	Integrate Bayesian neural networks or other probabilistic models to quantify epistemic and aleatoric uncertainty, enabling confidence-aware	Uncertainty modeling improves robustness and reliability, especially in data-scarce or noisy urban contexts, but remains underutilized (Nezhadettehad et al.,	High

	under data scarcity or noise.	parking management.	2025).	
Scalability challenges in spatial-temporal modeling for large urban networks	Advanced spatial-temporal models (e.g., GCNs, Transformers) face computational bottlenecks when scaling to citywide parking networks, limiting real-time deployment.	Develop scalable architectures using zone partitioning, graph sampling, or efficient spectral methods (e.g., Graph Cosine Operator) to reduce computational costs while preserving accuracy.	Large-scale urban parking systems require scalable models; current approaches either sacrifice granularity or incur high computational costs (Sun et al., 2023) (Yin et al., 2024) (Zhang et al., 2024).	High
Underexplored heterogeneity of parking types and spatial contexts	Few studies differentiate between on-street, off-street, mechanical, or shared parking types, or account for their distinct spatial-temporal dynamics and external factor sensitivities.	Conduct comparative analyses and develop specialized models for different parking types, incorporating tailored external factors and spatial heterogeneity.	Parking types exhibit different usage patterns and external influences; ignoring this heterogeneity reduces model generalizability and accuracy (Zhao et al., 2024) (Wang & Li, 2024) (Balmer et al., 2021).	Medium
Limited transferability and domain adaptation across cities	Models trained on one city or region often perform poorly when applied elsewhere due to differing urban fabrics and external factor distributions.	Advance transfer learning and adversarial domain adaptation frameworks that align feature spaces and contextual information for cross-city parking demand prediction.	Addressing data scarcity in new areas and improving model generalizability requires robust domain adaptation, which is currently nascent (Shao et al., 2024) (Shao et al., 2021).	Medium
Lack of standardized benchmarks and evaluation metrics	Diverse datasets, prediction horizons, and metrics hinder cross-study comparisons and practical	Establish standardized datasets, benchmarks, and multi-metric evaluation protocols for parking demand	Standardization facilitates reproducibility, fair comparison, and accelerates progress toward deployable solutions (Inam et al., 2022)	Medium

	assessment of model performance.	prediction incorporating external factors and interpretability.	(Soumana et al., 2023).	
Insufficient exploration of external factor temporal granularity and lag effects	Most models use coarse temporal granularity for external factors and do not systematically analyze lagged effects or temporal evolution of external influences.	Investigate multi-scale temporal modeling of external factors, including lagged and cumulative effects, to improve prediction accuracy and interpretability.	Temporal dynamics of external factors (e.g., weather changes, event timing) critically affect parking demand but are often oversimplified (Correia et al., 2020) (Ye et al., 2023).	Medium
Limited integration of economic and policy factors in predictive models	Few models incorporate economic variables (e.g., pricing, elasticity) or policy impacts explicitly, despite their significant influence on parking behavior.	Integrate economic modeling and policy-aware features into parking demand prediction frameworks, leveraging explainability tools to quantify their effects.	Economic and policy factors shape parking utilization and demand elasticity, essential for proactive management but underrepresented in current models (Niu et al., 2023) (Li et al., 2024).	Medium

Overall Synthesis and Conclusion

The current body of literature collectively underscores the significant influence of external factors such as weather conditions, socioeconomic attributes, special events, and urban infrastructure on the accuracy and robustness of parking demand prediction models in urban environments. Weather variables—including temperature, precipitation, and sunlight—consistently emerge as critical predictors that capture temporal fluctuations in parking occupancy. Socioeconomic factors like GDP, population density, and parking pricing exert complex and often nonlinear effects on parking utilization patterns, demonstrating the importance of integrating diverse contextual data beyond historical usage alone. Additionally, event-related variables and policy interventions have been shown to significantly modify parking behavior, necessitating adaptive modeling approaches that reflect dynamic urban conditions.

From a methodological standpoint, deep learning architectures, particularly LSTM, GRU, Transformer models, and graph convolutional networks, dominate the landscape due to their superior ability to capture intrinsic spatial-temporal dependencies and complex

nonlinear relationships inherent in parking data. The integration of multi-source data—including points of interest, traffic speeds, pedestrian flows, and policy indicators—further enriches these models' predictive power. However, these sophisticated models often entail higher computational costs and can pose challenges for real-time deployment, especially in large-scale urban networks. To mitigate these constraints, strategies such as parking zone division, hybrid architectures, and attention mechanisms have been proposed to balance efficiency and accuracy.

Explainability techniques, especially SHAP values, have gained traction for elucidating feature importance and enhancing model transparency. Their application reveals nuanced insights into the contribution of external factors, enabling urban planners and policymakers to understand and trust model predictions. Nevertheless, the adoption of SHAP remains limited and predominantly post-hoc, with opportunities for deeper integration into adaptive and uncertainty-aware frameworks largely unexplored. Uncertainty quantification through Bayesian and physics-informed neural networks represents a promising avenue to improve model robustness and decision-making under data scarcity and noise but remains underrepresented in the literature.

Overall, while considerable progress has been made in incorporating external factors and advanced machine learning methods to improve parking demand forecasting, challenges persist in achieving model generalizability across diverse urban contexts, real-time responsiveness, and comprehensive uncertainty modeling. Future research would benefit from developing holistic, scalable frameworks that dynamically integrate evolving external influences, enhance interpretability, and balance computational demands, thereby supporting smarter and more sustainable urban parking systems.

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